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Using Elimination

Another method for solving systems involves eliminating a variable by combining the equations. This is called the **elimination method**.

Example 1

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Solve:
$$\begin{cases} -7x - 9y = 17 & \textcircled{1} \\ 3x + 9y = 3 & \textcircled{2} \end{cases}$$

Step 1

Add the equations to eliminate the y -variable.

$$\begin{array}{r} -7x - 9y = 17 \\ 3x + 9y = 3 \\ \hline -4x + 0 = 20 \end{array} \quad \left. \begin{array}{l} \\ \end{array} \right\} \textcircled{1} + \textcircled{2}$$

Step 2

Solve for x .

$$\begin{array}{r} -4x = 20 \\ x = -5 \end{array} \rightarrow \text{substitute into } \textcircled{1} \text{ or } \textcircled{2}$$

Step 3

Substitute -5 for x in one of the equations and solve for y .

$$\begin{array}{r} -7x - 9y = 17 \quad \textcircled{1} \\ -7(-5) - 9y = 17 \\ 35 - 9y = 17 \\ -9y = -18 \\ y = 2 \end{array}$$

The solution is $(-5, 2)$.

Check the solution in the other equation.

$$\begin{array}{r} 3x + 9y = 3 \\ 3(-5) + 9(2) = 3 \\ -15 + 18 = 3 \\ 3 = 3 \quad \text{True} \end{array}$$

You may need to multiply one or more of the equations before combining. Remember, your goal is to get two terms that, when added, will equal zero.

Example 2

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Solve:
$$\begin{cases} -5x + 6y = -8 & \textcircled{1} \times 2 \\ 9x - 4y = 28 & \textcircled{2} \times 3 \end{cases}$$
 like finding LCD $6 \times 2 = 12, 4 \times 3 = 12$

Step 1

Choose which variable to eliminate. This example multiplies both equations so the y -coefficients equal 12 and -12 .

Step 2

Multiply every term in the first equation by 2.

$$\begin{array}{r} -5x + 6y = -8 \\ 2(-5x + 6y) = 2(-8) \\ -10x + 12y = -16 \end{array}$$

Multiply every term in the second equation by 3.

$$\begin{array}{r} 9x - 4y = 28 \\ 3(9x - 4y) = 3(28) \\ 27x - 12y = 84 \end{array}$$

Step 3

Add the resulting equations and solve for x .

$$\begin{array}{r} -10x + 12y = -16 \\ 27x - 12y = 84 \\ \hline 17x + 0 = 68 \end{array} \quad \left. \begin{array}{l} \\ \end{array} \right\} \textcircled{1} + \textcircled{2} \quad \text{watch signs}$$

$$\begin{array}{r} 17x = 68 \\ x = 4 \end{array} \rightarrow \text{into } \textcircled{1} \text{ or } \textcircled{2}$$

Step 4

Substitute 4 for x in one of the original equations and solve for y .

$$\begin{array}{r} -5x + 6y = -8 \\ -5(4) + 6y = -8 \\ -20 + 6y = -8 \\ 6y = 12 \\ y = 2 \end{array}$$

The solution is $(4, 2)$. Check the solution in the other equation.

$$\begin{array}{r} 9x - 4y = 28 \\ 9(4) - 4(2) = 28 \\ 36 - 8 = 28 \\ 28 = 28 \quad \text{True} \end{array}$$

or
for x:
 $5 \times 9 = 45$
 $9 \times 5 = 45$

Tip

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Good test takers use the problem-solving method that seems best for the situation. Use substitution when one of the variables will be easy to isolate. Use elimination when each variable has a number before it.

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